MODULE 7: WATER TECHNOLOGY AND CORROSION ENGINEERING

(CO: 7)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BCHY101L Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus): (1) Water hardness: temporary vs permanent, units (ppm, °Fr, °Clark), (2) EDTA complexometric titration mechanism, (3) Boiler troubles: scale, sludge, caustic embrittlement, and priming/foaming, (4) External water softening: Zeolite process and Ion-Exchange demineralization, (5) Desalination by Reverse Osmosis (RO), (6) Chemical and electrochemical corrosion: Hydrogen evolution and Oxygen absorption, (7) Differential aeration corrosion (waterline and pitting), and (8) Corrosion control: Cathodic protection (Sacrificial Anode & Impressed Current) and Galvanizing vs Tinning.

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1 VIT FAT Exam Blueprint & Master Formula Sheet for Module 7

In VIT Final Assessment Tests, Module 7 carries 20 to 25 marks:

1. **Question Type A (10 Marks):** Hardness calculations using EDTA titration data in ppm and °French, or Zeolite softener regeneration capacity.
2. **Question Type B (10 Marks):** Corrosion Engineering: Mechanism of wet electrochemical corrosion (H_2 evolution vs O_2 absorption) and differential aeration corrosion (waterline and pitting).
3. **Question Type C (10 Marks):** Water Treatment: Ion-Exchange demineralization process with service/regeneration reactions, or Reverse Osmosis membrane desalination.

Topic / Process	Governing Equation / Chemical Reaction	FAT Exam Significance
CaCO_3 Equivalents	$\frac{\text{Mass of salt (mg/L)} \times 50}{\text{Eq. Wt}}$	Benchmark: $Ca(HCO_3)_2 = 81, CaCl_2 = 55.5, MgSO_4 = 60$
EDTA Titration	1 mL 0.01 M EDTA $\equiv$ 1 mg $CaCO_3$	EBT indicator: wine red to steel blue at pH 10
Caustic Embrittlement	$Na_2CO_3 + H_2O \rightarrow 2NaOH + CO_2 \uparrow$	Stress-corrosion cracking at boiler rivets by localized concentration cell
Ion-Exchange Resins	$2R - SO_3H + Ca^{2+} \rightarrow (R - SO_3)_2Ca + 2H^+$	Produces 100% demineralized water; regenerated by 5% HCl and 5% NaOH
Wet Oxygen Absorption	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	Dominates rusting in neutral/alkaline aerated water

Galvanizing (Zn on Fe)	$E^0\left(\frac{Zn^{2+}}{Zn}\right) = -0.76$ V $< E^0\left(\frac{Fe^{2+}}{Fe}\right) = -0.44$ V	Sacrificial anode protection; protects iron even when deeply scratched

2 Topic 1: Water Hardness & Complexometric Titrations

2.0.1 Hardness Calculations & Equivalents

Water hardness is expressed in terms of calcium carbonate ($CaCO_3$) equivalents because $CaCO_3$ has a convenient molecular weight of 100 and equivalent weight of 50.

$$\text{CaCO}_3 \text{ equivalent} = \frac{\text{Mass of salt (mg/L)} \times 50}{\text{Equivalent Weight of salt}}$$

- **Conversion of Units:**

$$1 \text{ ppm} = 1 \text{ mg/L} = 0.1°\text{French} = 0.07°\text{Clark}$$

2.0.2 EDTA Titration Mechanism

At pH 9–10 (buffered with $NH_4OH + NH_4Cl$), Eriochrome Black T (EBT) forms an unstable wine-red complex with metal ions. Titration with disodium EDTA displaces the indicator, forming a stable colorless $[M - \text{EDTA}]^{2-}$ chelate and releasing free steel-blue indicator:

$$[\text{M-EBT}]^- (\text{Wine Red}) + H_2Y^{2-} \rightarrow [\text{M-Y}]^{2-} (\text{Colorless}) + HEBT^{2-} (\text{Steel Blue}) + H^+$$

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3 Topic 2: Boiler Troubles & External Softening

3.0.1 Boiler Troubles

1. **Scale and Sludge:** Sludge is loose and slimy (formed by $MgCO_3, CaCl_2$); scale is hard and adherent (formed by $CaSO_4$, silicates). Scale reduces thermal conductivity, leading to boiler plate explosion.
2. **Caustic Embrittlement:** Intercrystalline cracking at boiler rivets caused by alkaline concentration cells:

$$\text{Fe} \mid \text{Concentrated NaOH (Anode - Rivets)} \parallel \text{Dilute NaOH (Cathode)} \mid \text{Fe}$$

3.0.2 External Softening by Ion-Exchange (Demineralization)

- Cation resin ($R - SO_3H$) replaces all cations (Ca^{2+}, Mg^{2+}, Na^+) with H^+.
- Anion resin ($R - OH$) replaces all anions (Cl^-, SO_4^{2-}) with OH^-.
- $H^+ + OH^- \rightarrow H_2O$. Demineralized water has conductivity $< 0.1 \mu$S/cm.

4 Topic 3: Electrochemical Corrosion & Protection

4.0.1 Wet Electrochemical Corrosion

- **Anode (Oxidation):** Metal dissolves: $Fe \rightarrow Fe^{2+} + 2e^-$.
- **Cathode (Reduction):**
  - Acidic medium: $2H^+ + 2e^- \rightarrow H_2 \uparrow$ (Hydrogen evolution).
  - Neutral / aerated medium: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$ (Oxygen absorption).

4.0.2 Cathodic Protection Engineering

1. **Sacrificial Anode:** Attaching blocks of active metals (Mg, Zn) to steel structures. The active metal corrodes, keeping steel cathodic.
2. **Galvanizing vs Tinning:** Zinc coating (Zn) protects iron sacrificially even when scratched because $E^0(\text{Zn}) = -0.76 \text{ V} < E^0(\text{Fe}) = -0.44 \text{ V}$. Tin coating (Sn) protects iron only when intact; if scratched, iron corrodes rapidly because $E^0(\text{Sn}) = -0.14 \text{ V} > E^0(\text{Fe})$.

5 VIT Faculty Exam Insights & Common Traps

- **Galvanizing vs Tinning Rule:** Galvanized iron is used for roofing sheets and water pipes (sacrificial protection), but NEVER for food cans (zinc is toxic). Tinned iron is non-toxic and ideal for food containers.

6 Quick Self-Test Checklist Before Your Exam

1. Can you calculate Total, Temporary, and Permanent hardness in ppm and °Fr?
2. Write the complete ion-exchange service and regeneration reactions.
3. Why does iron undergo waterline corrosion just below the meniscus?
4. What is the electrochemical reason galvanized iron protects even when scratched?